

The Kardar–Parisi–Zhang equation

From growing fronts to universality

A colony boundary is irregular because local growth events fluctuate. A useful theoretical question is whether its large-scale roughness forgets the microscopic growth rules. The KPZ equation gives one possible answer. This note treats a one-dimensional interface in a two-dimensional setting, usually called 1 + 1 dimensions.

1 Why a slope-squared term appears

Let $h(x, t)$ be the single-valued interface height. For local growth, reflection symmetry in x , and invariance under a uniform height shift, a leading long-wavelength equation is

$$\partial_t h = v_0 + \nu \partial_x^2 h + \frac{\lambda}{2} (\partial_x h)^2 + \eta(x, t), \quad \nu > 0. \quad (1)$$

The coefficient ν smooths curvature, λ measures slope-dependent growth, and v_0 is a constant drift. The zero-mean Gaussian noise is idealized as

$$\langle \eta(x, t) \eta(x', t') \rangle = 2D \delta(x - x') \delta(t - t'), \quad D > 0. \quad (2)$$

There is no equilibrium fluctuation–dissipation relation assumed between D and ν . If x and h both have length units, ν has units of length squared per time, λ of length per time, and D of length cubed per time.

For a front moving at constant normal speed v_n , its vertical speed is $v_n \sqrt{1 + (\partial_x h)^2}$. Expanding for small slopes produces $v_n + v_n (\partial_x h)^2 / 2$. This explains the allowed nonlinearity geometrically; a coarse-grained model need not have $\lambda = v_0$. The average speed also receives a contribution from slope fluctuations.

2 The observable is a scaling law

On a periodic interval of length L , remove the instantaneous mean height $\bar{h}(t) = L^{-1} \int_0^L h \, dx$ and define

$$W^2(L, t) = \left\langle \frac{1}{L} \int_0^L [h(x, t) - \bar{h}(t)]^2 dx \right\rangle. \quad (3)$$

Starting from a flat front, dynamic scaling takes the form, up to nonuniversal scale factors,

$$W(L, t) \sim L^\alpha f(t/L^z), \quad f(y) \sim \begin{cases} y^\beta, & y \ll 1, \\ \text{constant}, & y \gg 1, \end{cases} \quad \beta = \alpha/z. \quad (4)$$

Thus the growth regime has $W \sim t^\beta$, while the saturation time scales as L^z . An exponent extracted from a short time interval can instead describe a crossover or a finite-resolution artifact.

3 Two routes to the one-dimensional exponents

For the standard periodic KPZ equation, the stationary distribution of height differences, with the mean fixed, is

$$P_{\text{st}}[h] \propto \exp \left[-\frac{\nu}{2D} \int_0^L (\partial_x h)^2 dx \right]. \quad (5)$$

Its slopes are spatial white noise subject to the zero-mean-slope constraint, so height increments have Brownian scaling: $\alpha = 1/2$. In one dimension the KPZ nonlinearity leaves this stationary measure invariant. It nevertheless changes the dynamics.

After subtracting the constant drift, the equation also has statistical tilt symmetry. On an infinite interface, for constant slope u ,

$$h_u(x, t) = h(x + \lambda u t, t) + u x + \frac{\lambda u^2 t}{2} \quad (6)$$

obeys the same equation with translated noise. At the nonlinear KPZ fixed point this symmetry gives $\alpha + z = 2$. Combining the relations yields

$$\boxed{\alpha = \frac{1}{2}, \quad z = \frac{3}{2}, \quad \beta = \frac{1}{3}.} \quad (7)$$

By contrast, setting $\lambda = 0$ gives the Edwards–Wilkinson equation, with $\alpha = 1/2$ but $z = 2$ and $\beta = 1/4$ in one dimension. Equal-time roughness alone cannot distinguish the two dynamics. These exact KPZ exponents should not be carried over unchanged to a two-dimensional interface.

4 Connections to other theories, and to biology

Differentiating Eq. (1) and writing $u = \partial_x h$ gives a noisy viscous Burgers equation,

$$\partial_t u = \nu \partial_x^2 u + \lambda u \partial_x u + \partial_x \eta. \quad (8)$$

For smooth regularized noise and with v_0 removed, the Cole–Hopf variable $Z = \exp[\lambda h/(2\nu)]$ formally obeys

$$\partial_t Z = \nu \partial_x^2 Z + \frac{\lambda}{2\nu} \eta Z. \quad (9)$$

This connects interface growth to diffusion in a random environment and, through a path integral, to directed polymers. In the space–time white-noise limit, $(\partial_x h)^2$ is singular: the Cole–Hopf construction involves an Itô stochastic heat equation and a renormalized height drift. The smooth-noise chain rule is not a complete definition of that limit.

For a microbial front, local reproduction with short-range fluctuations can motivate a KPZ description. Nutrient-mediated nonlocal growth, quenched obstacles, mechanical coupling, or overhangs can invalidate it. A useful comparison checks spatial roughness, temporal growth, and finite-size scaling together; the geometry and initial condition also affect fluctuation distributions [2].

5 Detailed benchmark: solve the linear problem first

The Edwards–Wilkinson limit is exactly soluble and provides a useful normalization and numerical check. Take $\lambda = 0$, a flat initial condition, and periodic length L . For $q = 2\pi n/L$, use

$$h_q(t) = \frac{1}{L} \int_0^L h(x, t) e^{-iqx} dx, \quad \langle \eta_q(t) \eta_{q'}(t') \rangle = \frac{2D}{L} \delta_{q, -q'} \delta(t - t'). \quad (10)$$

Remove the $q = 0$ mode, which translates the mean height. Each remaining Fourier mode solves an Ornstein–Uhlenbeck equation:

$$\dot{h}_q = -\nu q^2 h_q + \eta_q, \quad h_q(t) = \int_0^t e^{-\nu q^2(t-s)} \eta_q(s) ds, \quad (11)$$

$$\langle |h_q(t)|^2 \rangle = \frac{2D}{L} \int_0^t e^{-2\nu q^2(t-s)} ds = \frac{D}{\nu L q^2} (1 - e^{-2\nu q^2 t}). \quad (12)$$

Parseval's identity now gives the full width, not only its exponent:

$$W^2(L, t) = \frac{DL}{2\pi^2\nu} \sum_{n=1}^{\infty} \frac{1 - e^{-8\pi^2 n^2 \nu t / L^2}}{n^2}, \quad W^2(L, \infty) = \frac{DL}{12\nu}. \quad (13)$$

Both signs of q are included in this formula; counting them again would produce an erroneous factor of two. In the intermediate regime between a microscopic cutoff time and the finite-size time, replace the sum by an integral:

$$W^2 \simeq \frac{D}{\pi\nu} \int_0^\infty \frac{1 - e^{-2\nu q^2 t}}{q^2} dq = \frac{D}{\nu} \sqrt{\frac{2\nu t}{\pi}}. \quad (14)$$

The integral follows by differentiating with respect to t , evaluating a Gaussian integral, and integrating back from $t = 0$. Hence $W \propto t^{1/4}$ and the relaxation of the smallest mode gives a time proportional to L^2/ν .

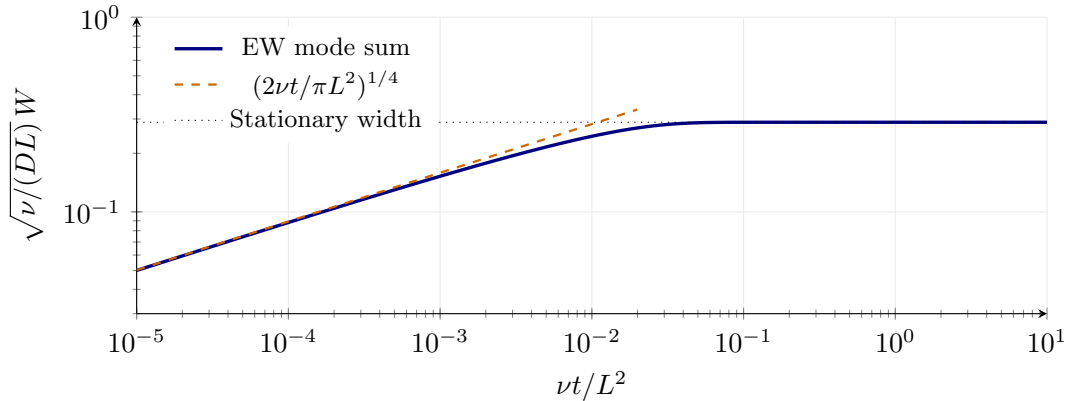


Figure 1: Exact EW mode sum, its early-time asymptote, and its saturation value. This is a linear-theory benchmark, not a simulated KPZ curve.

In a simulation, recovering both the amplitude and the crossover in Eq. (13) is a stronger check than fitting a straight line on logarithmic axes.

6 Why the nonlinear equation has the same stationary roughness

Let $\mathcal{P}[h, t]$ be the probability functional for the mean-subtracted field. Formally its Fokker–Planck equation is

$$\partial_t \mathcal{P} = - \int dx \frac{\delta}{\delta h(x)} \left[\left(\nu h_{xx} + \frac{\lambda}{2} h_x^2 \right) \mathcal{P} - D \frac{\delta \mathcal{P}}{\delta h(x)} \right]. \quad (15)$$

The spatially constant drift is projected out. For the Gaussian stationary candidate in the overview,

$$\frac{\delta \mathcal{P}_{\text{st}}}{\delta h(x)} = \frac{\nu}{D} h_{xx}(x) \mathcal{P}_{\text{st}}. \quad (16)$$

This cancels the diffusive probability current pointwise. The nonlinear contribution is divergence free, rather than zero. Its nontrivial field-dependent part contains

$$\int_0^L h_x^2 h_{xx} dx = \frac{1}{3} \int_0^L \partial_x (h_x^3) dx = 0. \quad (17)$$

The remaining functional-divergence term vanishes in a compatible symmetric periodic regularization. These are formal continuum manipulations: a spatial discretization must preserve the relevant integration-by-parts property to inherit this measure exactly. A zero divergence of probability current does not imply detailed balance; KPZ can retain a circulating probability current in configuration space.

The stationary slope covariance on a ring is consequently

$$\langle h_x(x) h_x(y) \rangle = \frac{D}{\nu} \left[\delta(x - y) - \frac{1}{L} \right]. \quad (18)$$

Integrating twice, for $0 \leq r \leq L$, gives

$$\langle [h(x+r) - h(x)]^2 \rangle = \frac{D}{\nu} r \left(1 - \frac{r}{L} \right). \quad (19)$$

The linear dependence at $r \ll L$ proves $\alpha = 1/2$. Integrating this increment variance over r and dividing by $2L$ reproduces $W^2 = DL/(12\nu)$ independently of the Fourier calculation.

Scaling, tilt symmetry, and a crossover estimate

Under $x \mapsto bx$, $t \mapsto b^z t$, $h \mapsto b^\alpha h$, the linear smoothing coefficient scales as $\nu \mapsto b^{z-2} \nu$ and the nonlinear coefficient as $\lambda \mapsto b^{z+\alpha-2} \lambda$. Tilt symmetry protects the nonlinear vertex in the standard KPZ theory; scale invariance with a nonzero nonlinear coupling therefore requires $z + \alpha = 2$.

At the EW fixed point, $z = 2$ and $\alpha = 1/2$, so the nonlinearity grows under coarse graining. A dimensionless coupling at length ℓ is

$$g(\ell) = \frac{D\lambda^2\ell}{\nu^3}, \quad \ell_{\text{cross}} \sim \frac{\nu^3}{D\lambda^2}, \quad t_{\text{cross}} \sim \frac{\nu^5}{D^2\lambda^4}. \quad (20)$$

These are weak-coupling scaling estimates, not universal numerical thresholds. If the sample is shorter than ℓ_{cross} , an experiment may saturate before reaching the KPZ regime.

7 Cole–Hopf in detail, and what to measure

First use a smooth noise field so ordinary calculus applies, subtract $v_0 t$, and set $c = \lambda/(2\nu)$ and $Z = e^{ch}$. Direct differentiation gives

$$Z_t = cZ \left(\nu h_{xx} + \frac{\lambda}{2} h_x^2 + \eta \right), \quad (21)$$

$$Z_{xx} = cZ h_{xx} + c^2 Z h_x^2. \quad (22)$$

Because $\nu c^2 = c\lambda/2$, the slope-squared terms cancel in $Z_t - \nu Z_{xx}$, leaving $Z_t = \nu Z_{xx} + c\eta Z$. Linearity in Z does not make the problem deterministic: the potential multiplying Z is random.

For a spatially smoothed noise that is white in time, define dW_ϵ by

$$\langle dW_\epsilon(x, t) dW_\epsilon(y, t) \rangle = 2DR_\epsilon(x - y)dt, \quad (23)$$

where R_ϵ approaches a spatial delta function as the smoothing length tends to zero. Starting from the Itô stochastic heat equation

$$dZ = \nu Z_{xx}dt + cZ dW_\epsilon, \quad (24)$$

Itô's formula for $h = c^{-1} \ln Z$ gives

$$dh = \left[\nu h_{xx} + \frac{\lambda}{2} h_x^2 - cDR_\epsilon(0) \right] dt + dW_\epsilon. \quad (25)$$

The last drift diverges as the spatial cutoff is removed. It is the explicit counterterm in this regularization. Absorbing the divergent mean growth into the height reference is part of defining continuum KPZ, not an optional correction to an otherwise ordinary product of distributions.

For smooth noise and flat positive initial Z , the Feynman–Kac representation expresses Z as an average of $\exp[c \int \eta(B_s, t - s) ds]$ over Brownian paths B_s with generator $\nu \partial_x^2$. This is a directed-polymer partition sum; $h = c^{-1} \ln Z$ is proportional to its random free energy, with a sign set by conventions. The exact distribution of fluctuations also depends on flat, curved, or stationary initial data, even when the growth exponent agrees.

A practical sequence for a biological front

Define a single-valued front and a consistent mean subtraction before fitting. Measure the increment variance versus separation, the width versus time, and saturation versus system size. Use several fitting windows to expose crossovers. Test whether noise is approximately short-range in space and time, and whether nutrient transport creates a nonlocal growth law. A power law in one observable alone is weak evidence for a universality class.

For numerical work, check Eq. (12) first. A cell of width Δx has Euler noise increment $\sqrt{2D\Delta t/\Delta x}$. Explicit diffusion requires Δt of order $\Delta x^2/\nu$ or smaller; the nonlinear term imposes further constraints. Check convergence under grid refinement, not only numerical stability.

References

- [1] M. Kardar, G. Parisi, and Y.-C. Zhang, “Dynamic Scaling of Growing Interfaces,” *Phys. Rev. Lett.* **56**, 889 (1986). [doi:10.1103/PhysRevLett.56.889](https://doi.org/10.1103/PhysRevLett.56.889).
- [2] K. A. Takeuchi, *An appetizer to modern developments on the Kardar–Parisi–Zhang universality class*. [arXiv:1708.06060](https://arxiv.org/abs/1708.06060).